



Figure 1: Structure of a block in the chain

▪ Immutability of data

The immutability of the data which sits on the blockchain is perhaps the most powerful and convincing reason to deploy blockchain-based solutions for a variety of socio-economic processes that are currently recorded on centralised servers. This 'unchanging over time' feature makes the blockchain useful for accounting and financial transactions, in identity management and in asset ownership, management and transfer, just to name a few examples. Once a transaction is written onto the blockchain, no one can change it or, at least, it would be extremely difficult to do so.

▪ Transparency

Transparency of data is embedded within the network as a whole. The blockchain network exists in a state of consensus, one that automatically checks in with itself. Due to the structure of a block, the data in a blockchain cannot be corrupted; hence altering any unit of information in it is almost impossible. Though, in theory, this can be done by using a huge amount of computing power to override the entire network, this is not possible practically.

▪ Decentralisation

By design, the blockchain is a decentralised technology. Anything that happens on it is a function of the network, as a whole. A global network of computers uses blockchain technology to jointly manage the database that records transactions. The consensus mechanism discussed next ensures the correctness of data stored on the blockchain.

▪ Security

By storing data across its network, the blockchain eliminates the risks that come with data being held centrally, and the network lacks centralised points of vulnerability that are prone to being exploited. The blockchain ensures all participants in the network use encryption technologies for the security of the data. Primarily, it uses PKI (public key infrastructure), and it is up to the participants to select other encryption technologies as per their preference.

What are consensus mechanisms and the types of consensus algorithms?

Consensus is an agreement among the network peers; it refers to a system of ensuring that participants agree to a certain state of the system as the true state. It is a process whereby the peers synchronise the data on the blockchain. There are a

number of consensus mechanisms or algorithms. One is *Proof of Work*. Others include *Proof of Stake*, *Proof of Elapsed Time* and *Simplified Byzantine Fault Tolerance*. Bitcoin and Ethereum use *Proof of Work*, though Ethereum is moving towards *Proof of Stake*.

What are smart contracts?

Back in 1996, a man named Nick Szabo coined the term 'smart contract'. You can think of it as a computer protocol used to facilitate, verify, or enforce the negotiation of a legal contract. A smart contract is a phrase used to describe computer code. Smart contracts are simply computer programs that execute predefined actions when certain conditions within the system are met. Smart contracts provide the language of transactions that allows the ledger state to be modified. They can facilitate the exchange and transfer of anything of value (e.g., shares, money, content and property).

Open source blockchain frameworks, projects and communities

Looking at the current state of research and some of the implementations of blockchain technologies, we can certainly say that most enterprise blockchain initiatives are backed by open source projects. Here's a list of some of the popular open source blockchain projects.

- **Hyperledger** is an open source effort created to advance cross-industry blockchain technologies. Hosted by the Linux Foundation, it is a global collaboration of members from various industries and organisations.
- **Quorum** is a permissioned implementation of Ethereum, which supports data privacy. Quorum achieves data privacy by allowing data visibility on a need-to-know basis, using a voting-based consensus algorithm. Interestingly, Quorum was created and open sourced by J.P. Morgan.
- **Chain Core**, created by *chain.com*, was initially designed for financial services institutions and for things like securities, bonds and currencies.
- **Corda** is a distributed ledger platform designed to record, manage and automate legal agreements between businesses. It was created by the R3 Company, which is a consortium of over a hundred global financial institutions. 

References

- [1] <https://www.hyperledger.org/>
- [2] <https://github.com/jpmorganchase/quorum/wiki>
- [3] <https://chain.com/>
- [4] <https://www.corda.net/>

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